S. B. JAIN INSTITUTE OF TECHNOLOGY, MANAGEMENT & RESEARCH, NAGPUR.

Practical No. 8

Aim: Demonstrate the concept of Semantic network on imputed predicate logic to represent the knowledge.

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AIM: Demonstrate the concept of Semantic network on imputed predicate logic to represent the knowledge.

OBJECTIVE/EXPECTED LEARNING OUTCOME:

The objectives and expected learning outcome of this practical are:

- Understand semantic networks as a knowledge representation technique
- Implement ISA and HAS-A relationships
- Demonstrate inheritance in semantic networks
- Query knowledge programmatically using Python

THEORY:

Semantic Network

A **semantic network** is a graphical knowledge representation technique consisting of:

- **Nodes** → Concepts / Objects
- **Edges** → Relationships

Common relations:

- ISA → Inheritance
- HAS-A → Property
- CAN → Ability

It is closely related to **predicate logic representation**, where:

isa(dog, animal).
has(dog, tail).
can(bird, fly).

Relationship to Predicate Logic

A semantic network can be represented using first-order predicates:

Semantic Relation	Predicate Form
Dog is an Animal	isa(dog, animal)
Bird can Fly	can(bird, fly)
Dog has Tail	has(dog, tail)

Inheritance rule:

If

isa(X, Y) and has(Y, Z)

Then

has(X, Z)

Semantic Network Example

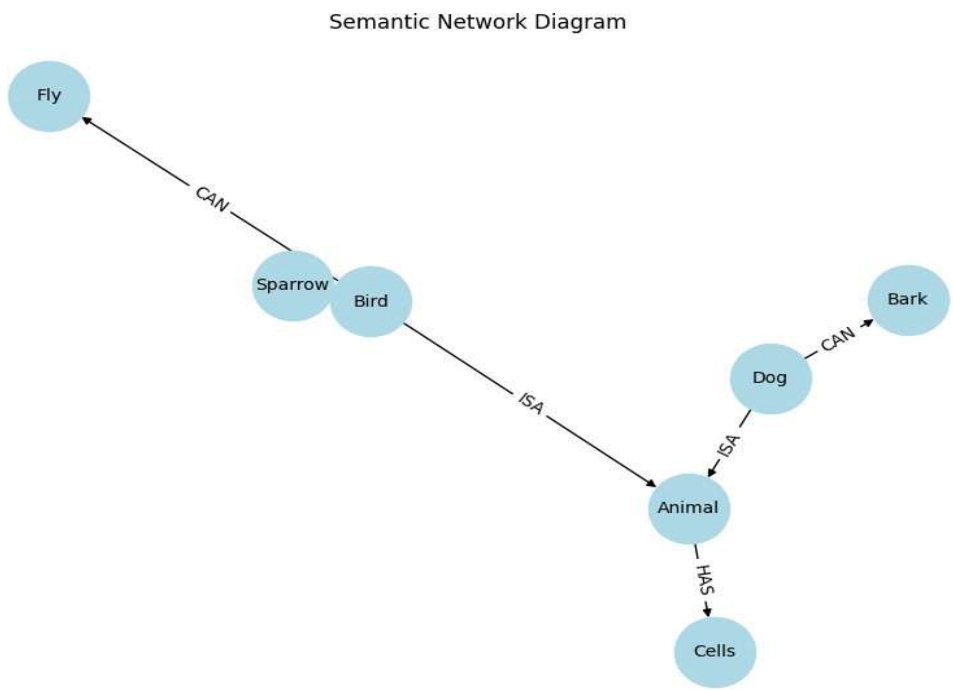
Nodes:

- Animal
- Bird
- Dog
- Sparrow

Relationships:

- Bird ISA Animal
- Dog ISA Animal
- Sparrow ISA Bird
- Animal has cells
- Bird can fly
- Dog can bark

SEMENTIC NETWORK DIAGRAM:



PROGRAM CODE:

main.py

```python
isa = {
    "Bird": "Animal",
    "Dog": "Animal",
    "Sparrow": "Bird"
}

has = {
    "Animal": "Cells"
}

# CAN relationship (Ability)
can = {
    "Bird": "Fly",
    "Dog": "Bark"
}

# Function to check ISA relationship
def check_isa(x, y):
    while x in isa:
        if isa[x] == y:
            return True
        x = isa[x]
        return True
    while x in isa:
        x = isa[x]
        if x in has and has[x] == property:
            return True
    return False

# Function to check CAN ability with inheritance
def check_can(x, ability):
    if x in can and can[x] == ability:
        return True
    while x in isa:
        x = isa[x]
        if x in can and can[x] == ability:
            return True
    return False

print("Sparrow ISA Bird:", check_isa("Sparrow", "Bird"))
print("Dog ISA Animal:", check_isa("Dog", "Animal"))
print("Sparrow HAS Cells:", check_has("Sparrow", "Cells"))
print("Bird CAN Fly:", check_can("Bird", "Fly"))
print("Dog CAN Bark:", check_can("Dog", "Bark"))
```

INPUT & OUTPUT:

```
Sparrow ISA Bird: True
Dog ISA Animal: True
Sparrow HAS Cells: True
Bird CAN Fly: True
Dog CAN Bark: True

=== Code Execution Successful ===
```

CONCLUSION:

DISCUSSION QUESTIONS:

1. What is a semantic network?

2. What is ISA relationship?

3. What is inheritance in knowledge representation?

4. How is predicate logic used here?

5. What are advantages of semantic networks?

REFERENCES:

- *https://www.geeksforgeeks.org/artificial-intelligence/semantic-networks-in-artificial-intelligence/*
- *https://thecloudflare.com/knowledge-representation-using-the-semantic-network-in-ai/*

https://www.siliconvalleycloudit.com/knowledge-representation-in-ai-semantic-networks/